

Lab protocol Day 2 – gRNA cloning part 2 and donor generation

To check for colonies that have a correct gRNA insert in p201 we perform colony PCR to amplify over the insertion site and send the PCR product for sequencing. We also inoculate precultures to have sufficient cells from which to extract correctly assembled and sequence-perfect gRNA plasmids tomorrow.

We will also prepare the PTC donor DNA templates for the editing experiment.

Colony PCR and inoculation of miniprep cultures (2h):

Materials: transformation plate, 4 sterile toothpicks or tips, sterile water, 4 glass tubes, LB + spectinomycin, Sapphire Master Mix, o297 and o298 at 10 μ M, 12 PCR tubes, magnetic beads, agarose, TBE buffer, 4 Qubit tubes, Qubit dsDNA BR working solution, 4 sequencing tubes

Equipment: bucket with ice, thermocycler, mini tabletop centrifuge, magnetic stand for PCR tubes, electrophoresis equipment, gel imager, Qubit device

1. Pick a single colony into 50 μ L sterile water in a PCR tube. Transfer the rest of the same colony to a glass tube with 5 ml LB + spectinomycin. **Do for 4 colonies per plate.**
2. Boil PCR tube at 96°C for 10 min.
3. Spin down briefly in a mini tabletop centrifuge, then use the supernatant as template for the PCR reaction.
4. Mix the following components in a PCR tube to make a 1.1*x master mix (x=4) of all components but template.

	1x	4.4x
Sapphire 2X master mix	12.5 μ L	55
FWD primer, 10 μ M	1 μ L	4.4
REV primer, 10 μ M	1 μ L	4.4
sterile water	8.5 μ L	37.4
Template	2 μ L	

Colony PCR primers (length = 307 bp):

FWD: o297 5' aaggcagcggtatcatcaac 3'

REV: o298 5' acggcgtattcagatcctc 3'

5. Aliquot 23 μ L of the master mix into each PCR tube, add 2 μ L template, vortex + spin down.
6. Place tubes in thermocycler and run program.

Temperature	Time	Number of cycles
98°C	1 min	Single cycle
98°C	5 s	10 cycles
55.5°C	5 s	
72°C	5 s	
98°C	5 s	20 cycles
57.5°C	5 s	
72°C	5 s	
72°C	1 min	Single cycle

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7. While the PCR is running place the glass tubes in a 37°C incubator in a roller drum for overnight incubation.
8. Make 1x gel by mixing x g agarose and x mL TBE buffer (for 100 ml gel, use 1g agarose and 100 ml TBE)
9. Heat the mixture in the microwave until dissolved.
10. Let it cool down for 5 min.
11. Add x drops ethidium bromide (1 drop per 50 mL gel). Pour the gel and let it solidify for 30 minutes.
12. After the PCR, load 5 µL sample in each well of the gel (adding loading dye is not required as SapphireAmp mix already contains dye), add 5 µL SmartLadder in lane 1.
13. Connect tank to gel electrophoresis machine, set to 130 V, 150 W & 400 mA and run for 30 minutes.
14. Check for correct band under UV light.
15. Clean up the PCR product of positive samples (307 bp) using 1.5X ratio magnetic beads (same protocol as Day 1 but adjust bead volume).
16. Elute PCR products in 10 µL water.
17. Determine concentration using the Qubit device (same protocol as Day 1).
18. Add ~5 ng sample with 2.5 µL o297 (10 µM) and fill up with sterile water to 10 µL in sequencing tubes.
19. Write down number of sequencing kit + position of samples and fill them in on the Eurofins website (**done by Vonesch lab**).

Donor DNA generation via 1-cycle extension (40 min):

Materials: donor FWD and REV oligos at 100µM, 4 PCR tubes, 1 Eppendorf tube, Phusion polymerase and HF buffer, dNTPs, sterile water, ice cold 100% EtOH, 3M sodium acetate, 70% EtOH, 1 Qubit tube, Qubit dsDNA BR working solution

Equipment: bucket with ice, thermocycler, PCR tube holder, vortex, mini tabletop centrifuge, Qubit device

This step creates the double-stranded linear donor template for editing of PTCs. The donors for the whole gene KO are already made and provided by the Vonesch lab (see *SOEing PCR_protocol.pdf* for stepwise instructions).

Order your designed donor so that you have 50 bp of homology either side of the intended mutation; design as two complementary fragments overlapping for ~20 bp; check that melting temperature of overlap region is at least 48°C.

Donor 1 (groups 1 & 4)

FWD: o2502 5' ggcgctggcgcggtgaatttgattggtgaacacaccgactacaacgactgattcgttct 3'
REV: o2503 5' tgcacaactgatcacggttgataatcaatcgcgagggcagaacgaatcagtcgttgta 3'
T_{ANN, donor1} = 54 °C

Donor 2(groups 2 & 5)

FWD: o2504 5' aacagcttcggcgcggtggacatggtgatcagcggaatgtgccgcagtaagccggggtta 3'
REV: o2505 5' atacggttcggcgcgacttcagtgaaagcgaagaacttaaccgggcttactcgggca 3'
T_{ANN, donor2} = 65.5 °C

Donor 3(groups 3 & 6)

FWD: o2506 5' tgcgggatcatggatcagctaatttcgcgctcggcaagaagatcattgattgctgac 3'
REV: o2507 5' gggcatggaaactgcttgggtcccagtgagcggaatcgatcagcaatcaatgatctt 3'
T_{ANN, donor3} = 53 °C

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1. Mix the following components on ice and split into 4 PCR tubes:

sterile water	134	μL
10X HF buffer	20	μL
dNTPs	4	μL
Donor FWD oligo	20	μL
Donor REV oligo	20	μL
Phusion polymerase	2	μL

2. Place tubes into thermocycler and run the following program:

98°C 1min

T_{ANN} 30s

72°C 3 min

15°C 5 min

3. Pool reactions in an Eppendorf tube, add 2.5x sample volume ice cold 100% EtOH (from -80°C) and 1/10 sample volume 3M sodium acetate.
4. Incubate 20 min (longer is also ok) at -80°C
5. Spin 15 min at 4°C 15'000 rpm
6. Wash once with 1mL 70% EtOH, vortex briefly and spin down again for 5 min at 4°C 15'000 rpm.
7. Remove supernatant and let dry (2 - 5 min).
8. Resuspend in 35 μL sterile water (incubate at 37°C, optionally shaking, to dissolve faster).
9. Measure donor concentration using the Qubit device (same protocol as Day 1).

Inoculation of *E. coli* strain for editing (5 min):

Materials: plate with editing strain, 1 glass tube, LB + ampicillin

For editing, we will use strain BW25113 (genotype F- LAM- rpoS396(Am) rph-1. lacI+, Ara- and Rha-) already transformed with our editing plasmid p205 (Cas9 expressed from pTET promotor, Lambda Red beta/gam/exo expressed from arabinose-inducible pBAD promoter).

Select a single colony from the streak plate and inoculate in 3 mL LB + ampicillin. Grow overnight at 37°C in a roller drum.